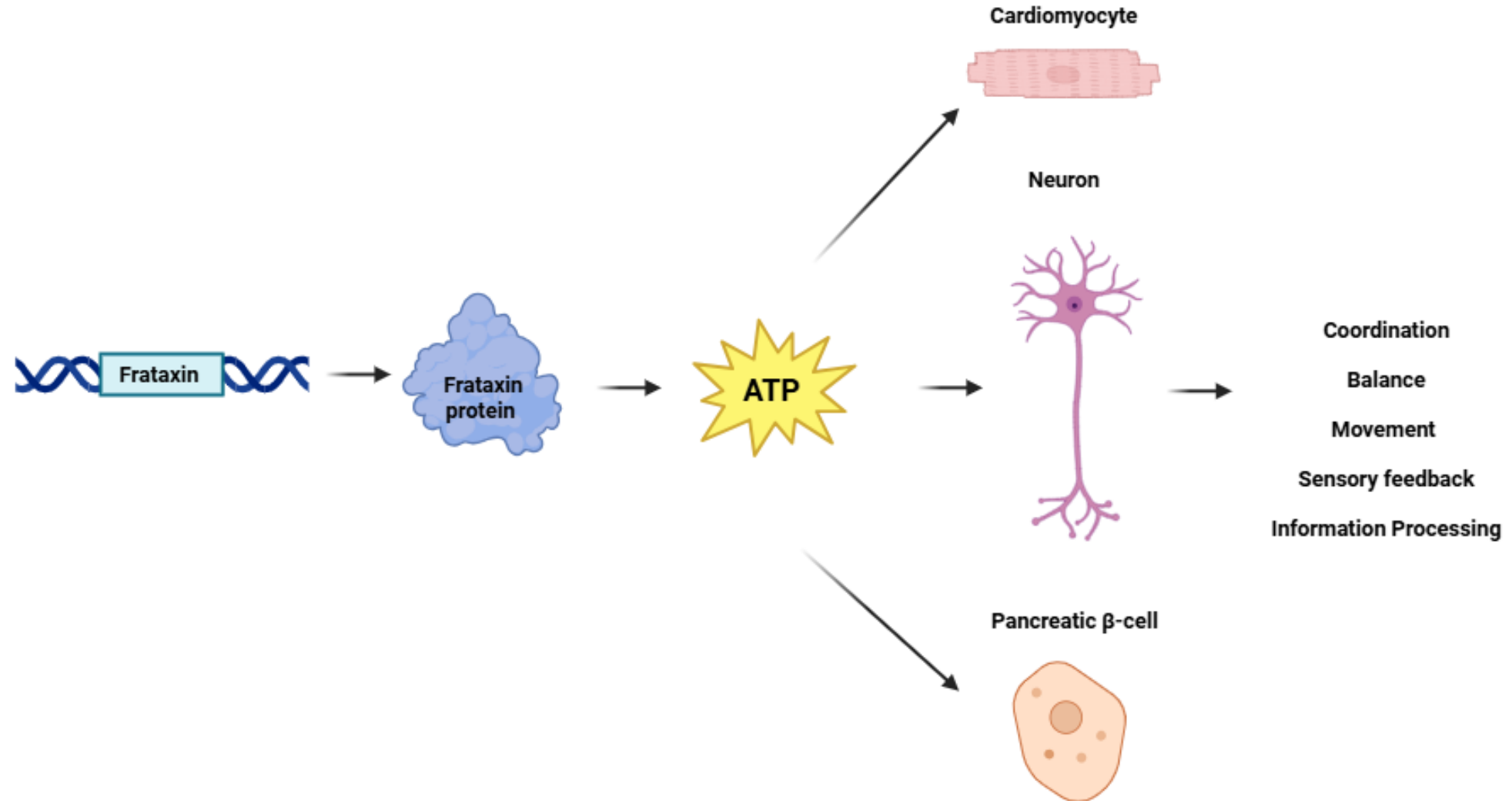




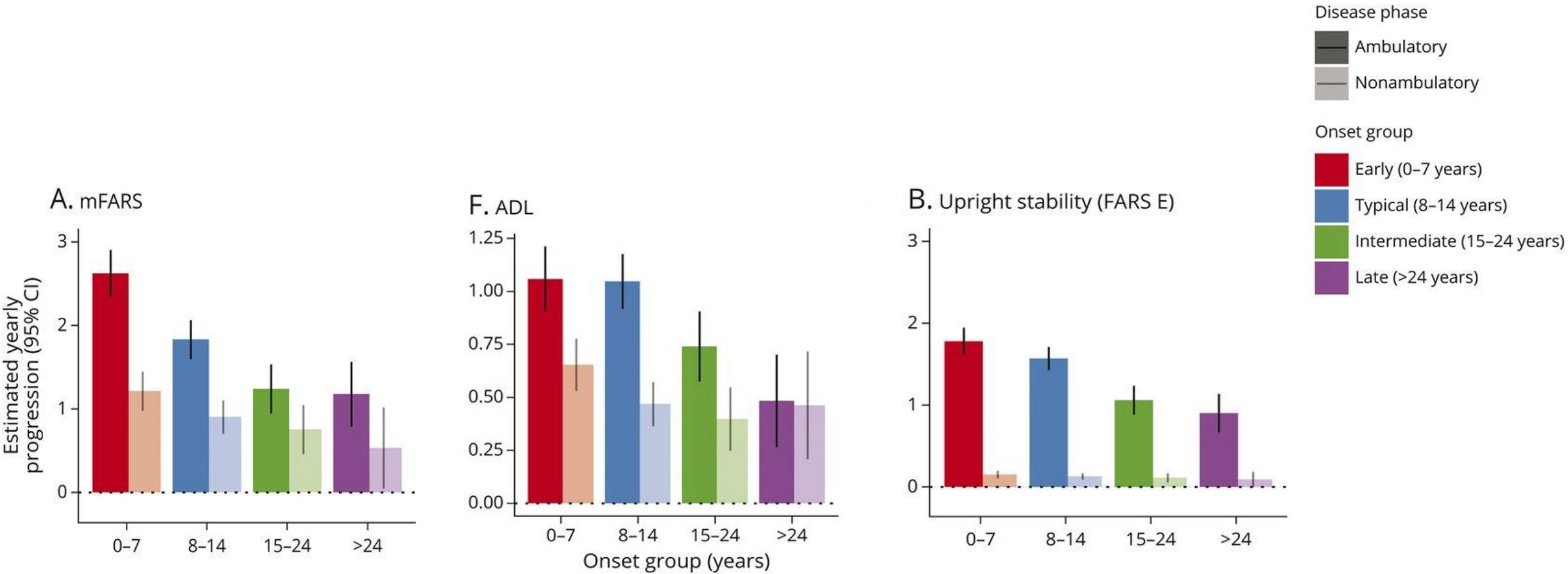
AtaxAI: Combating Neurodegeneration with Artificial Intelligence

Abby Micallef, Tamar Shalev, Rasik Sharma, Prabir Singh, and Dyan Socransky

Friedreich's Ataxia results from a GAA trinucleotide repeat expansion in the FXN gene, reducing frataxin production and ATP, ultimately impairing neuronal function



Symptoms vary from mobility issues to slurred speech, with faster progression observed in patients with earlier symptom onset due to longer FXN nucleotide expansions



The current Friedreich's Ataxia market of \$1.2 billion USD is comprised of one approved drug, omaveloxolone, and several treatments currently in trials

DRIVER	(~) % IMPACT ON CAGR FORECAST	GEOGRAPHIC RELEVANCE
First disease-modifying approval (omaveloxolone)	+3.2%	North America, Europe
Gene-therapy R&D ramp-up	+2.8%	North America, Europe, Asia-Pacific
Broader patient registries	+2.1%	Global, with APAC acceleration
Newborn-screen panel expansion	+1.9%	North America, Europe, selective APAC
AI-enabled drug repurposing	+1.6%	Global research hubs
Orphan-drug tax incentives	+1.4%	Primarily North America, growing in EU

<https://www.mordorintelligence.com/industry-reports/friedreichs-ataxia-market>

Integrating Artificial Intelligence To Build Complete Biocognitive Profiles For Patients With Friedrich's Ataxia

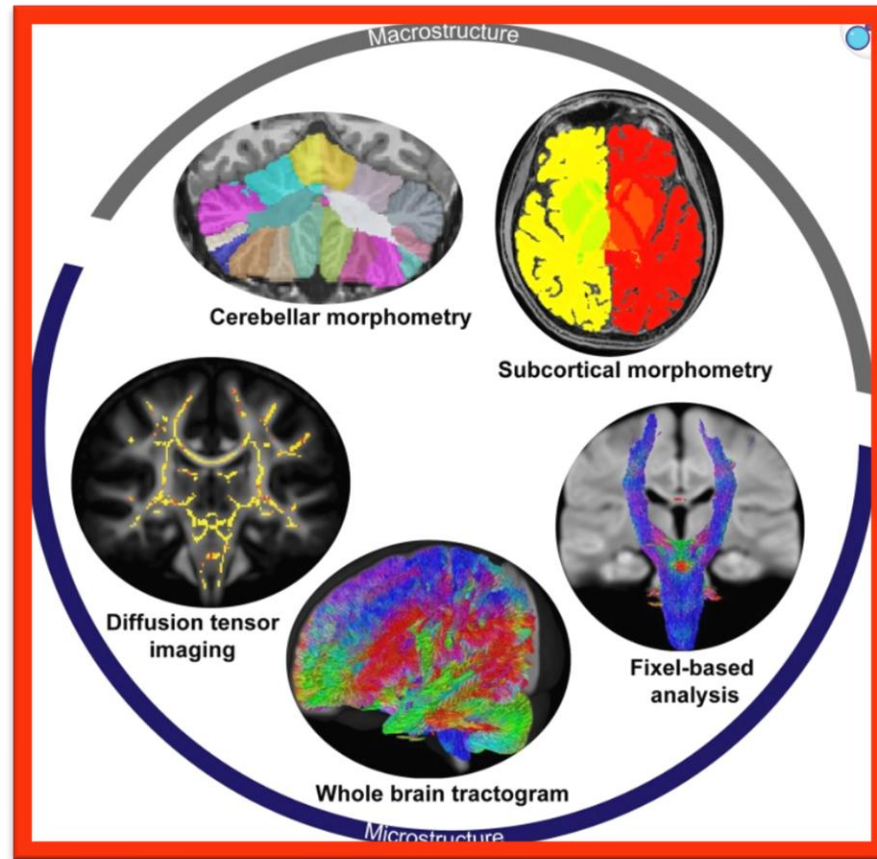


Figure F: Different brain imaging techniques that can visually display the progression of Friedrich's Ataxia [X]

- Purchasing imaging and patient data from healthcare networks
- Allowing patients to self-submit their records and previous imaging
- Administering cognitive and physical self-assessments to users for benchmark parameters

Patients can access AtaxAI consistently; daily use supports growth and targets personalized needs.

| Your
suggested
daily tasks



Matching
Activity



Up & Down
the Stairs



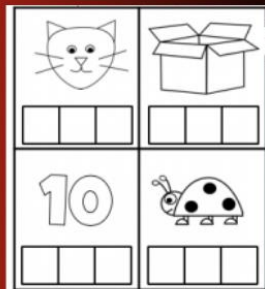
Balance
Activity



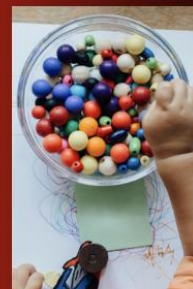
Learn more about
Friedreich's ataxia
and how to
support activities
of daily living



Counting



Identify
the Object



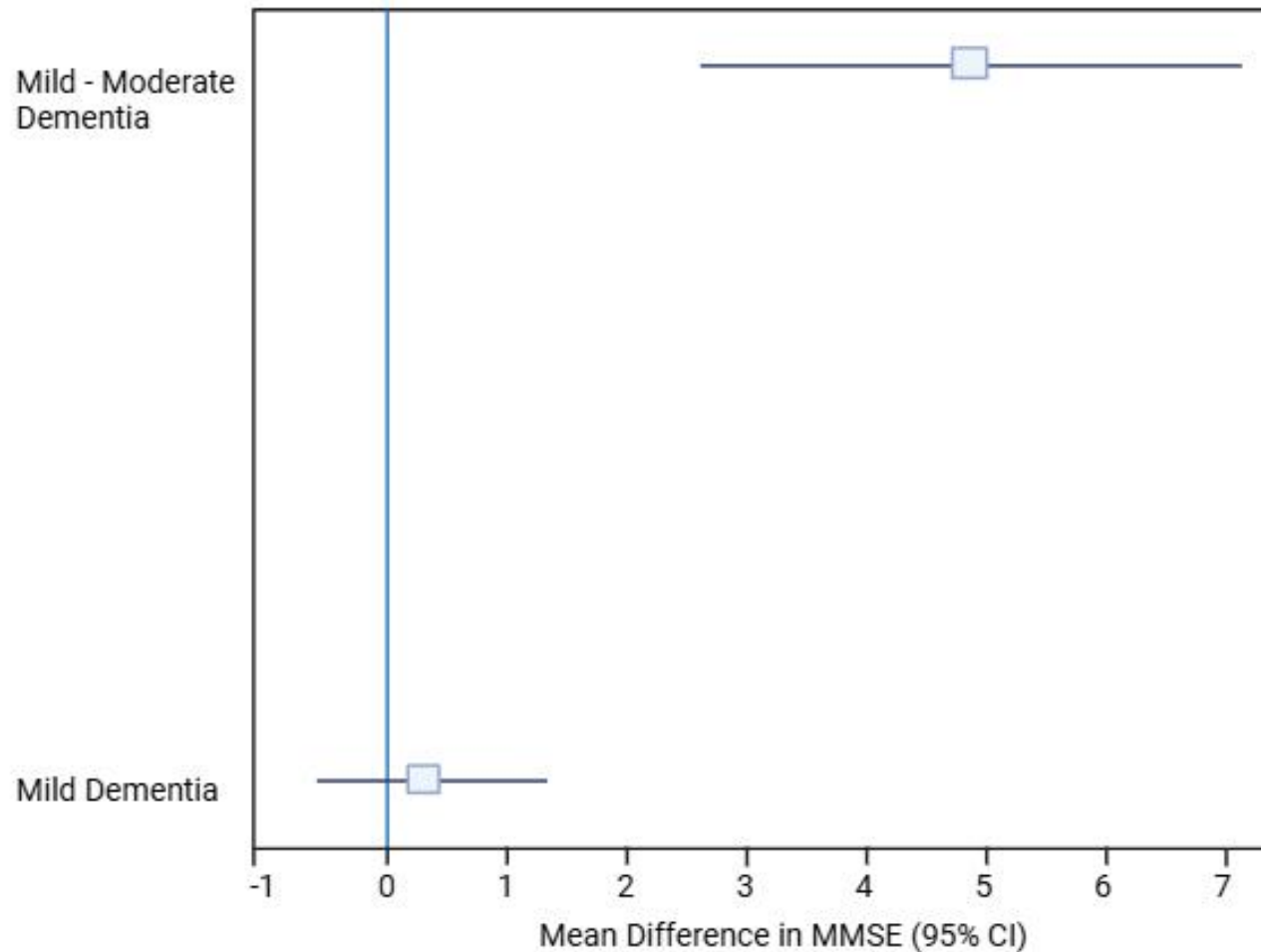
Fine Motor
Skills

Figure 4: Sample dashboard of a patient's suggested cognitive and physical daily tasks.

Cognitive Training Shows a Clear Benefit in Patients with Mild-Moderate Dementia.

Mild Dementia:
0.38 MMSE,
95%CI (-0.66 to
1.41)

Mild-Moderate
Dementia:
4.96MMSE,
95%CI (2.61 to
7.30)



<https://link.springer.com/article/10.1007/s1101400-z>

Physical rehabilitation significantly improves motor performance and ataxia symptoms in patients with FRDA – with significant reduction on the FARS and SARA scales.

	Pre (N = 42)	Post (N = 42)	Value of p^*
SARA	17.4 (7.8)	16.1 (7.9)	<0.001
FARS total	53.7 (17.2)	49.4 (17.7)	<0.001
<i>FARS LL</i>	<i>8.7 (4.4)</i>	<i>7.6 (4.5)</i>	<i><0.001</i>
<i>FARS UL</i>	<i>11.5 (4.6)</i>	<i>9.4 (4.2)</i>	<i><0.001</i>
<i>FARS US</i>	<i>18.5 (6.0)</i>	<i>17.4 (6.7)</i>	<i><0.001</i>

FARS – Friedrich's Ataxia Rating Scale

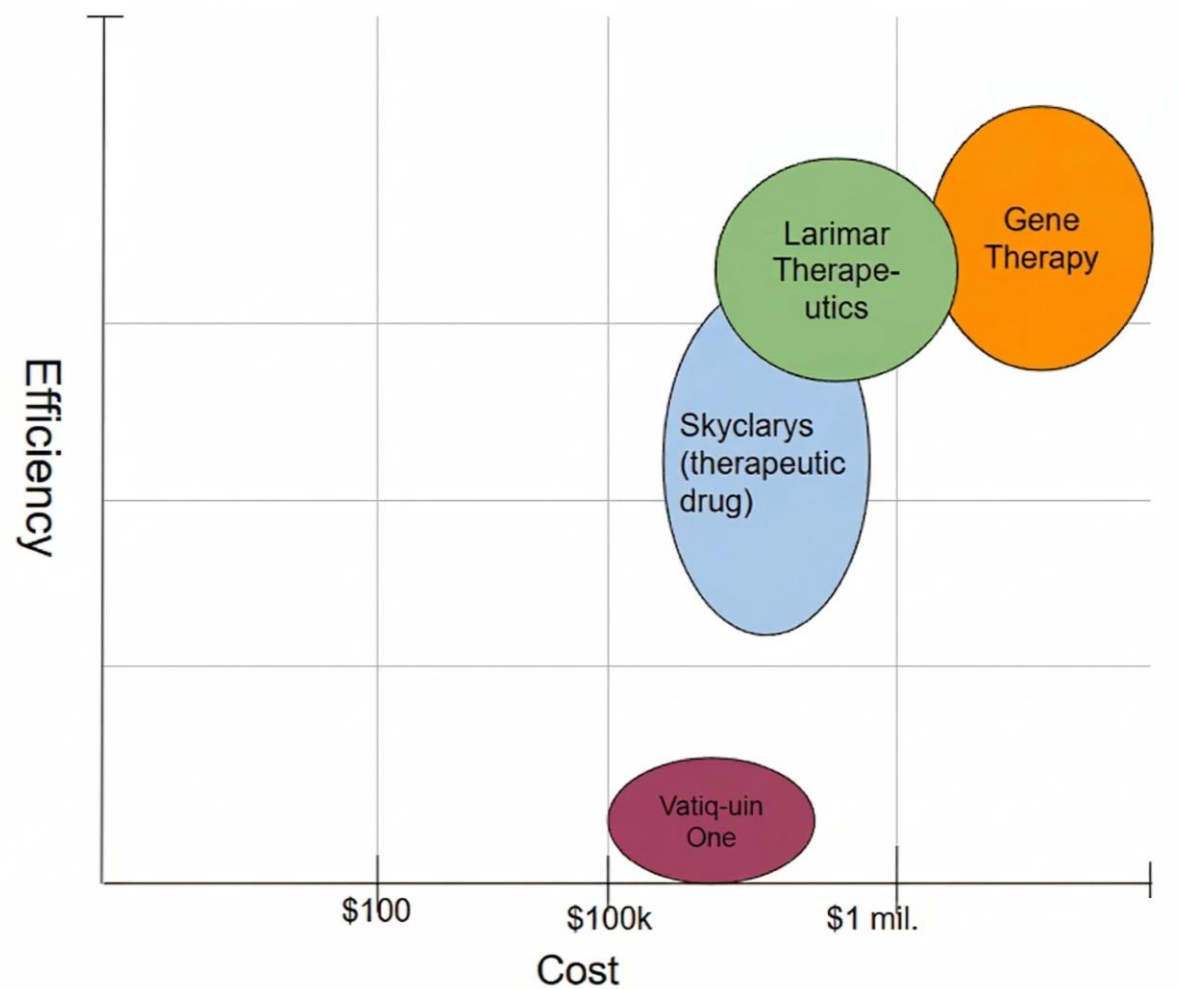
SARA – Scale for the Assessment and Rating of Ataxia

Figure 5: SARA and FARS scale measurements before and after physical rehabilitation.

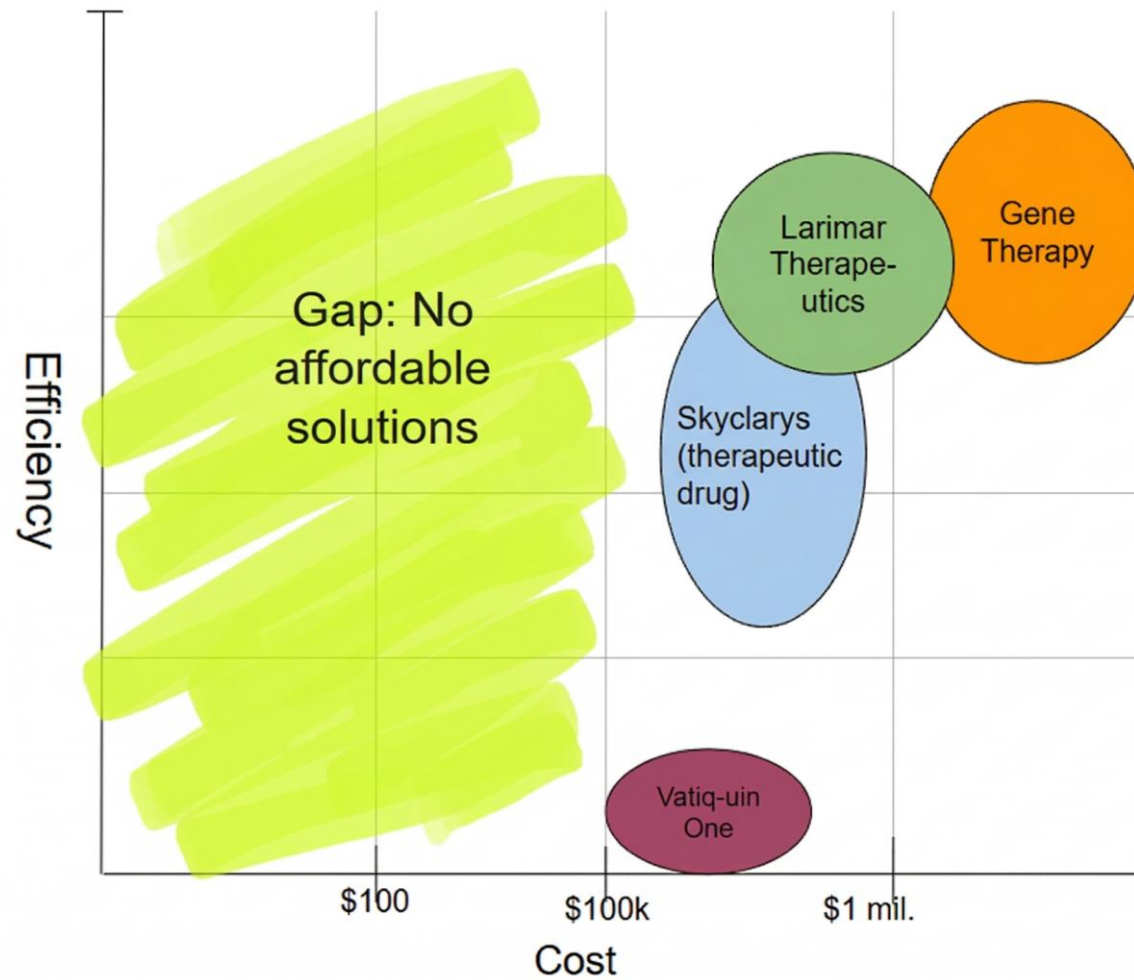
Competition

	Affordability	Personalization	Accessibility	Cognitive Outcomes	Physical Outcomes
Physiotherapy/ Rehab	✗	✓	✓	✗	✓
Scyclarys	✗	✗	✗	✓	✓
Gene Therapy	✗		✗		✓
AtaxAI	✓	✓	✓	✓	✓

Market Opportunity

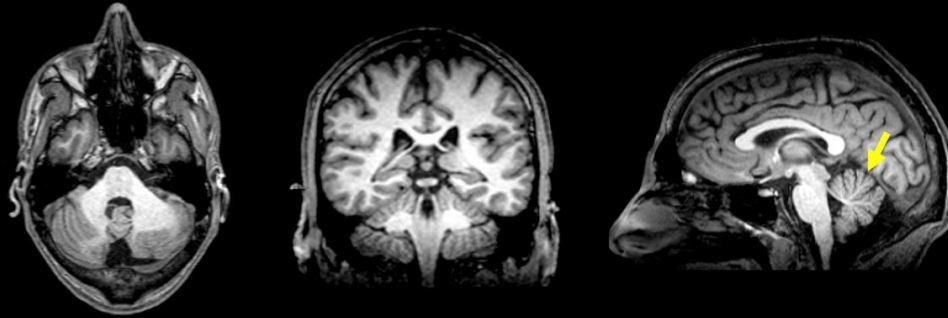


Market Opportunity



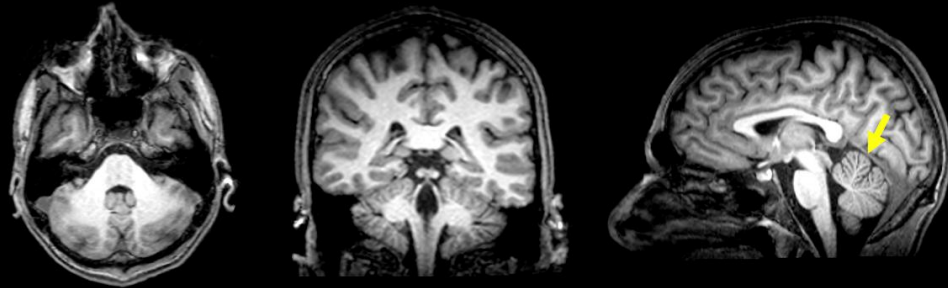
Technical Feasibility

A



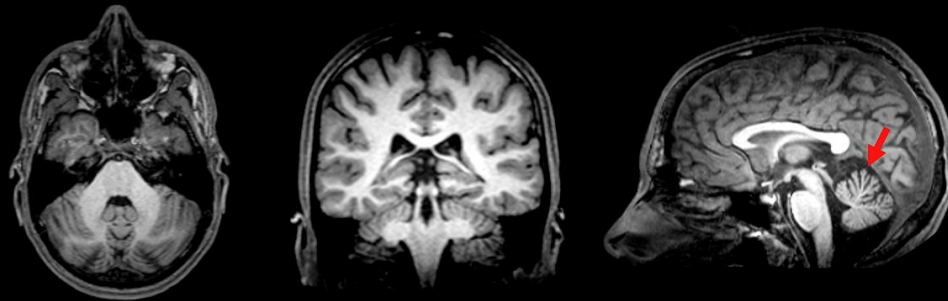
20-year-old male participant
Classical subtype
SuStaIn Stage: 15

B



18-year-old male participant
Early Cerebral subtype
SuStaIn Stage: 4

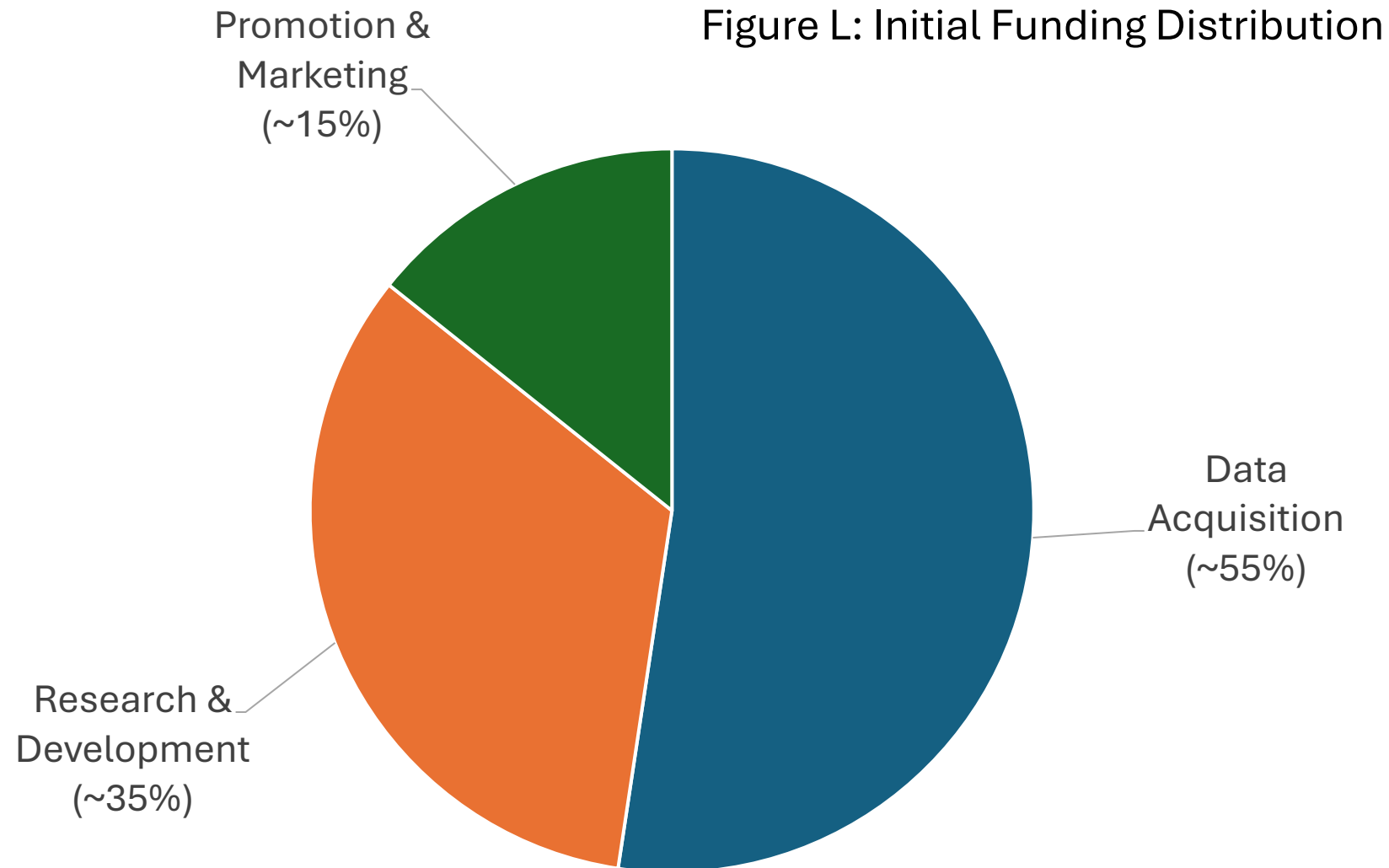
C



19-year-old male participant
Early Cerebellar subtype
SuStaIn Stage: 13

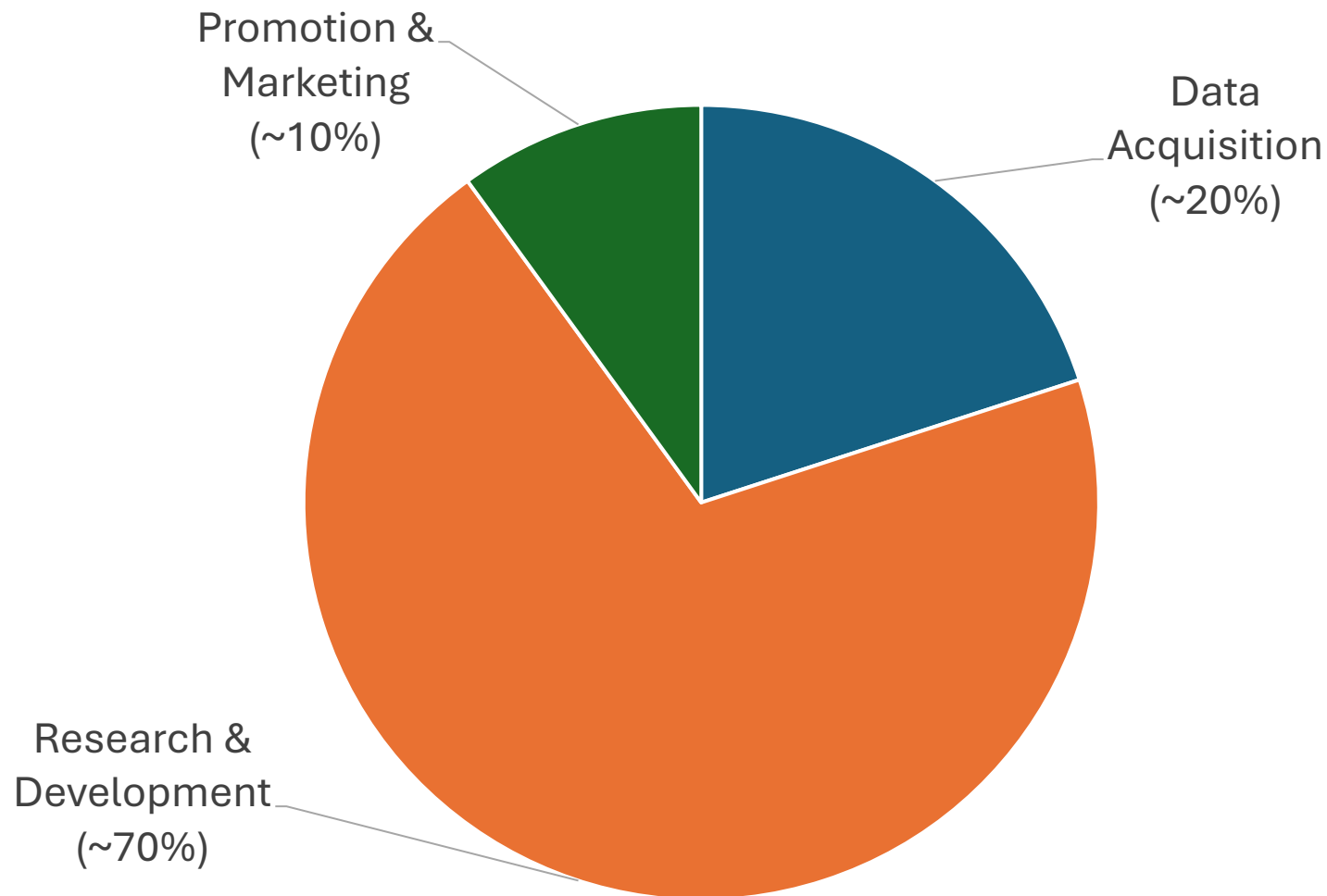
Placeholder

Initial funding allocation will include marketing towards clinics and hospitals, data purchases from imaging centres, and research and development



Over the long-term, funding will pivot towards research & development to investigate applications in other neurodegenerative contexts

Figure G: Long-term Funding Distribution



Utilizing external funding, AtaxAI can position itself as a disruptive innovator in the field of neurodegenerative disease



AtaxAI

References

X. Adanyeguh IM, Joers JM, Deelchand DK, Hutter DH, Eberly LE, Guo B, Iltis I, Bushara KO, Henry PG, Lenglet C. Brain MRI detects early-stage alterations and disease progression in Friedreich ataxia. *Brain Commun.* 2023 Jul 6;5(4):fcad196. doi: 10.1093/braincomms/fcad196. PMID: 37483529; PMCID: PMC10360047.

Appendix

Title